

#ASLI ENGINEERING

LFU with Approximate Counting

SWIPE

BY

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LFU and Approximate Counting

Apart from LRU, Redis also supports LFU eviction strategy

Core idea: Evict the key that is least frequently used

Every object in Redis has 24 bits for LRU, } space optimized
it piggybacks LFU on the same set of bits }

logarithmic Counter

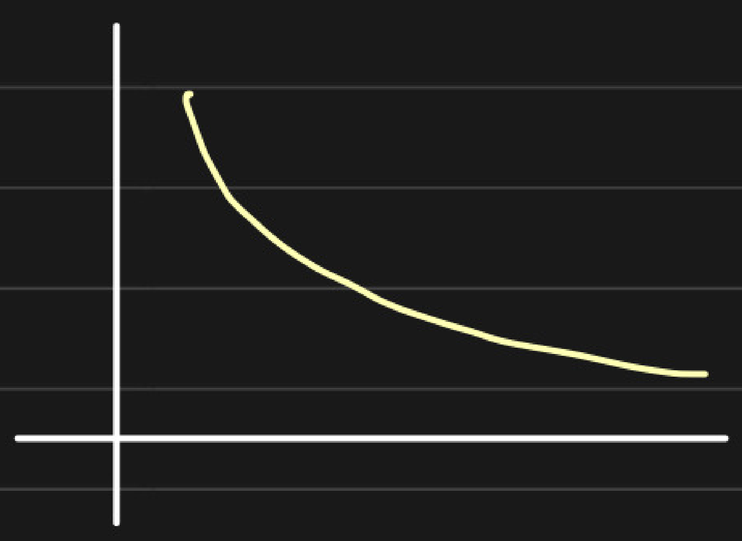
↳ approximate access frequency



* if not approximate, we would be able to count till $2^8 = 256$ only

Last Decrement Time

What if a key was heavily accessed in the past, but not accessed anymore?



The key would by lying around....

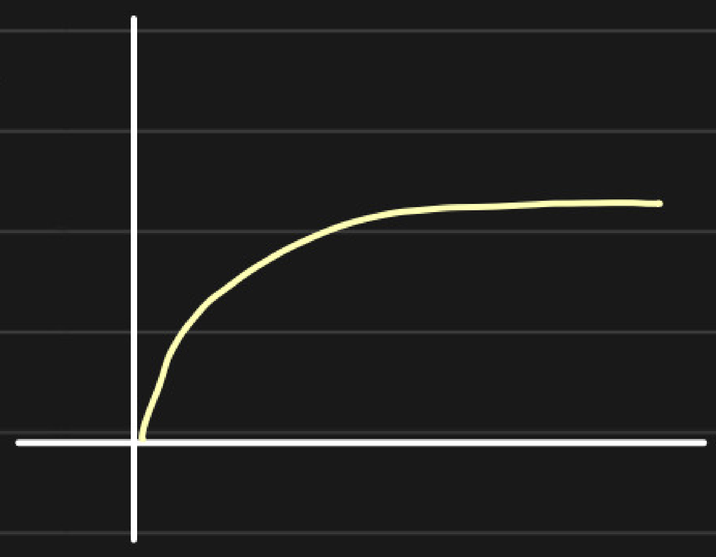
16 bits store Unix Time in Minutes when the key was last accessed.

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Morris Counter (logarithmic counter)

Instead of storing 'n' (actual value), lets store $v = \ln(1+n)$

highlight: v will grow faster for smaller value
and slower for larger one



n = 1
n = 2
n = 5
} even a small change is significant
"higher precision for smaller values"

n = 10000
n = 10100
n = 11000
} relative difference is not significant enough. We can lose precision for larger values

Incrementing the value

all we know is v and not n
↓ ↳ actual value
what we've stored

Challenge:

we cannot just increment v .... resultant value will rise exponentially.

hence, we need a good way to determine when to increase v

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